

# Applied NLP

DATA 641

Lecture 4 — Text Representation; One-hot encoding

\*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

# Vectorization: Big Idea

Let us start with a basic idea of text representation: map each word in the vocabulary ( $V$ ) of the text corpus to a unique ID (integer value), then represent each sentence, social media post, or document in the corpus as a  $V$ -dimensional vector. To better understand this let us take a toy corpus of four documents,  $D_1, D_2, D_3, D_4$ .

|    |                |
|----|----------------|
| D1 | Dog bites man. |
| D2 | Man bites dog. |
| D3 | Dog eats meat. |
| D4 | Man eats food. |

After lowercasing text and removing punctuation, the vocabulary of this corpus is comprised of six words: [dog, bites, man, eats, meat, food].

We can organize the vocabulary in any order. In this example, we simply take the order in which the words appear in the corpus. Every document in this corpus can now be represented with a vector of size six.

# One-hot encoding

We encode our corpus as a one-hot numeric array.

- One Hot Encoding: In one-hot encoding, each word  $w$  in corpus vocabulary is given a unique integer id  $w_{id}$  that is between 1 and  $|V|$ , where  $V$  is the set of corpus vocabulary. Each word is then represented by a  $V$ -dimensional binary vector of 0s and 1s.
- This is done via a  $|V|$  dimension vector filled with all 0s barring the index, where index =  $w_{id}$ . At this index, we simply put a 1. The representation for individual words is then combined to form a sentence representation.
- Example: D1: Dog bites man, D2: Man bites dog, D3: Dog eats meat, D4: Man eats food. We first map each of the six words to unique IDs: dog = 1, bites = 2, man = 3, meat = 4, food = 5, eats = 6. For D1, each word is a six-dimensional vector. Dog is represented as  $[1 \ 0 \ 0 \ 0 \ 0 \ 0]$ , as the word "dog" is mapped to ID 1. Thus, D1 is represented as  $\begin{bmatrix} [1 \ 0 \ 0 \ 0 \ 0 \ 0] & [0 \ 1 \ 0 \ 0 \ 0 \ 0] \\ [0 \ 0 \ 1 \ 0 \ 0 \ 0] \end{bmatrix}$ . Other documents in the corpus can be represented similarly.

Link for the official documentation of both can be found [here](#).

On the positive side, one-hot encoding is intuitive to understand and straightforward to implement. However, it suffers from the following:

- The size of a one-hot vector is directly proportional to size of the vocabulary, and most real-world corpora have large vocabularies. This results in a sparse representation where most of the entries in the vectors are zeroes, making it computationally inefficient to store, compute with, and learn from (sparsity leads to over-fitting).
- This representation does not give a fixed-length representation for text, i.e., if a text has 10 words, you get a longer representation for it as compared to a text with 5 words. For most learning algorithms, we need the feature vectors to be of the same length.
- Say we train a model using our toy corpus. At runtime, we get a sentence: "man eats fruits." The training data did not include "fruit" and there is no way to represent it in our model. This is known as the out of vocabulary (OOV) problem. A one-hot encoding scheme cannot handle this.